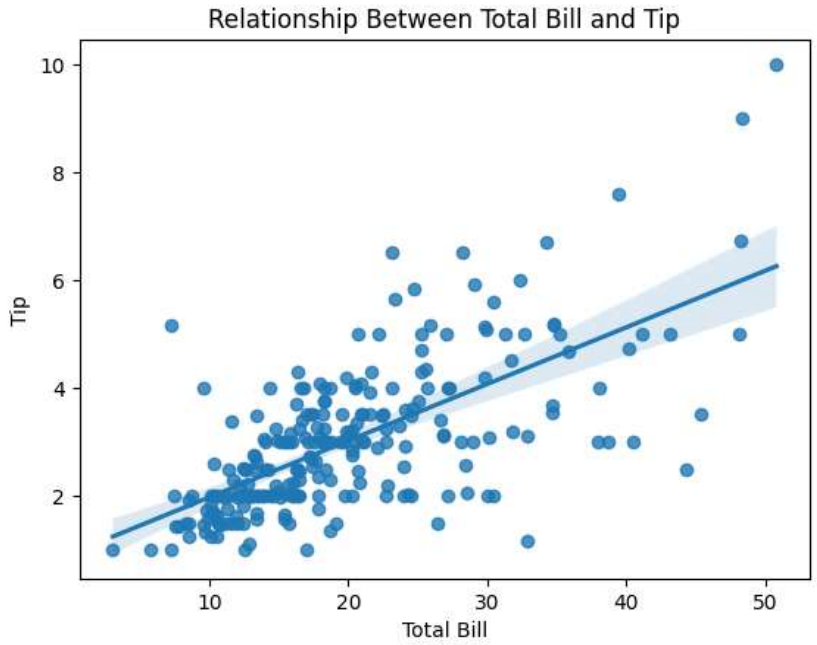
Manju Goyal
0801IT231073
B. Tech. 4th year - A3

LAB ASSIGNMENT - 03

Q1. Load the built-in 'tips' dataset in Seaborn and use regplot() to visualize the relationship between total_bill and tip, including the confidence interval band.

Code:

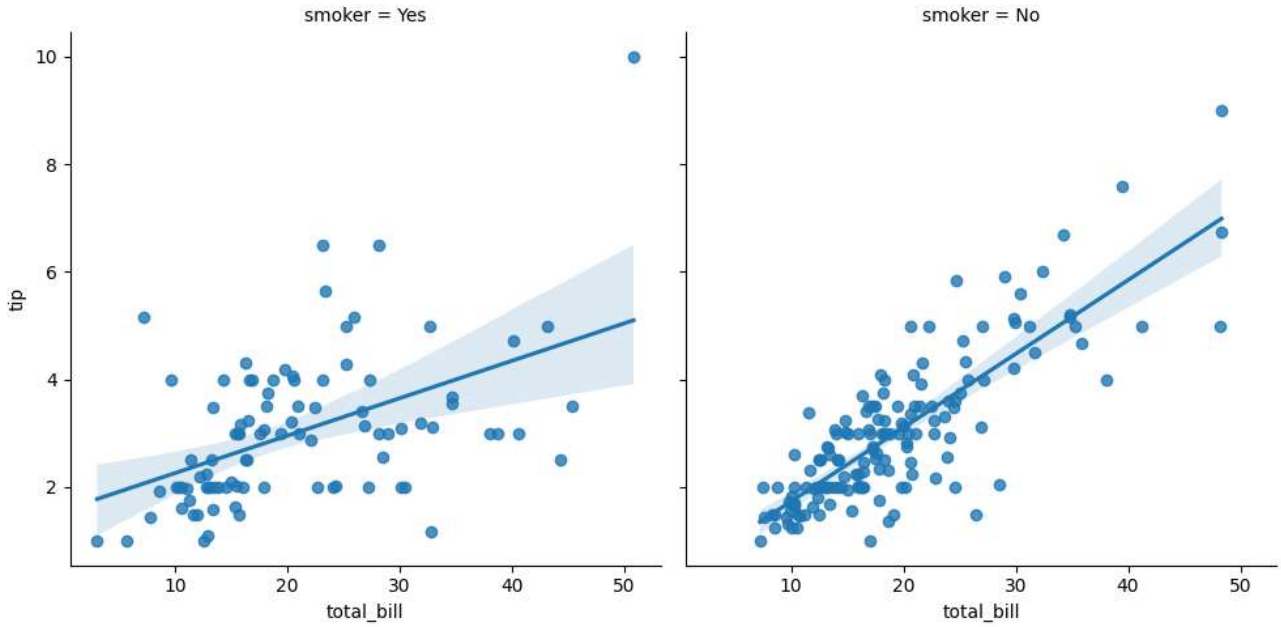
```
import seaborn as sns
import matplotlib.pyplot as plt
# Load tips dataset
df = sns.load_dataset("tips")
# Regression plot
sns.regplot(
    data=df,
    x="total_bill",
    y="tip"
)
plt.title("Relationship Between Total Bill and Tip")
plt.xlabel("Total Bill")
plt.ylabel("Tip")
plt.show()
```



Q2. Using the 'tips' dataset, create an lmplot() that shows a separate regression line for each value of the 'smoker' column, split across columns using the col parameter.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load tips dataset
df = sns.load_dataset("tips")
# Create regression plots for smokers and non-smokers
sns.lmplot(
    data=df,
    x="total_bill",
    y="tip",
    col="smoker"
)
plt.show()
```

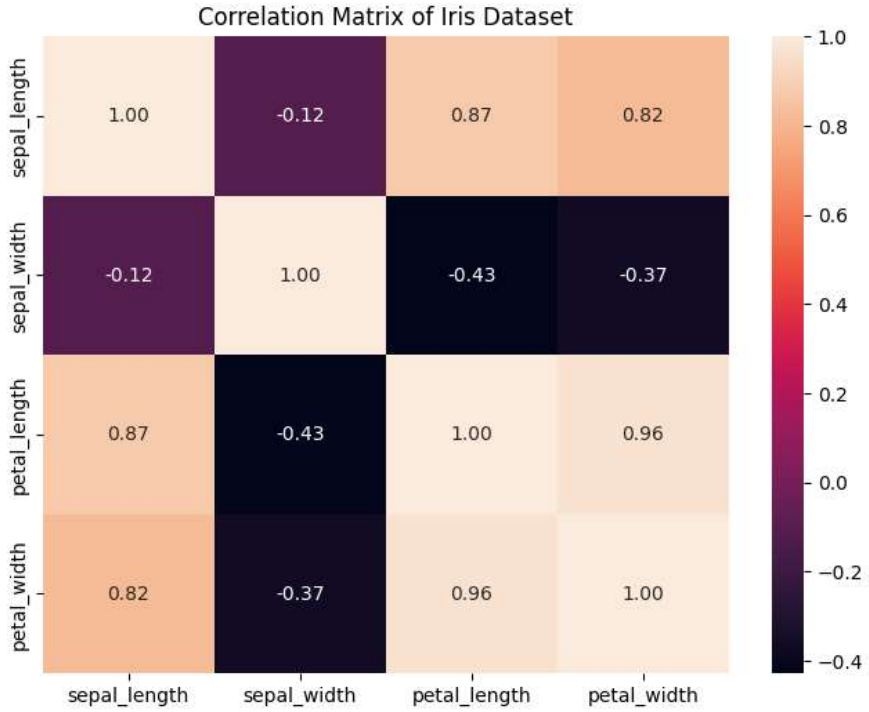


Q3. Take any dataset with at least four numeric columns and plot its correlation matrix using heatmap(), displaying the correlation values inside each cell.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load iris dataset
df = sns.load_dataset("iris")
```

```
# Select numeric columns
numeric_data = df.select_dtypes(include="number")
# Calculate correlation matrix
correlation = numeric_data.corr()
# Create heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f"
)
plt.title("Correlation Matrix of Iris Dataset")
plt.show()
```

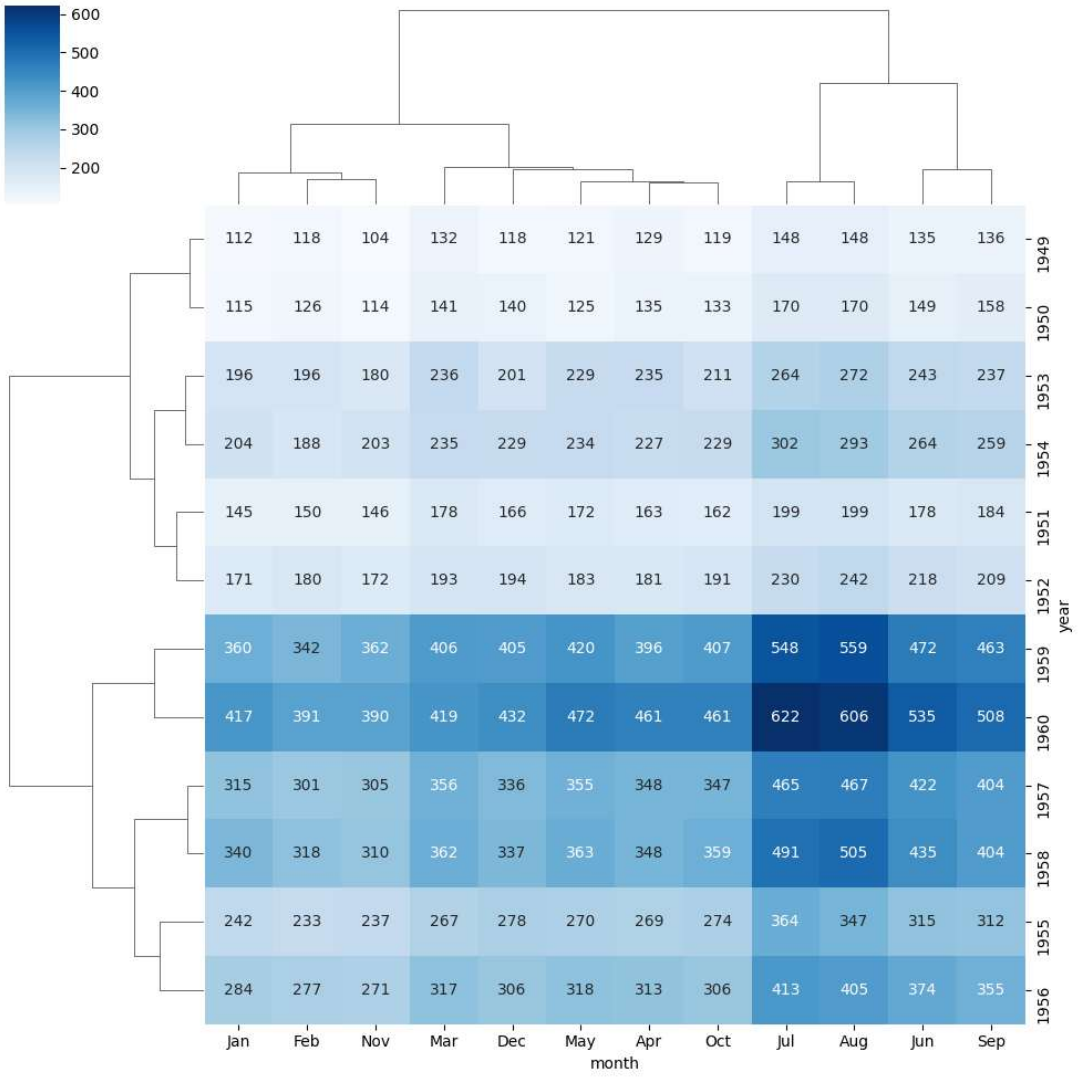


Q4. Use the 'flights' dataset (pivoted into a matrix of years vs months) and plot it using clustermap() to show hierarchical clustering of rows and columns.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load flights dataset
df = sns.load_dataset("flights")
```

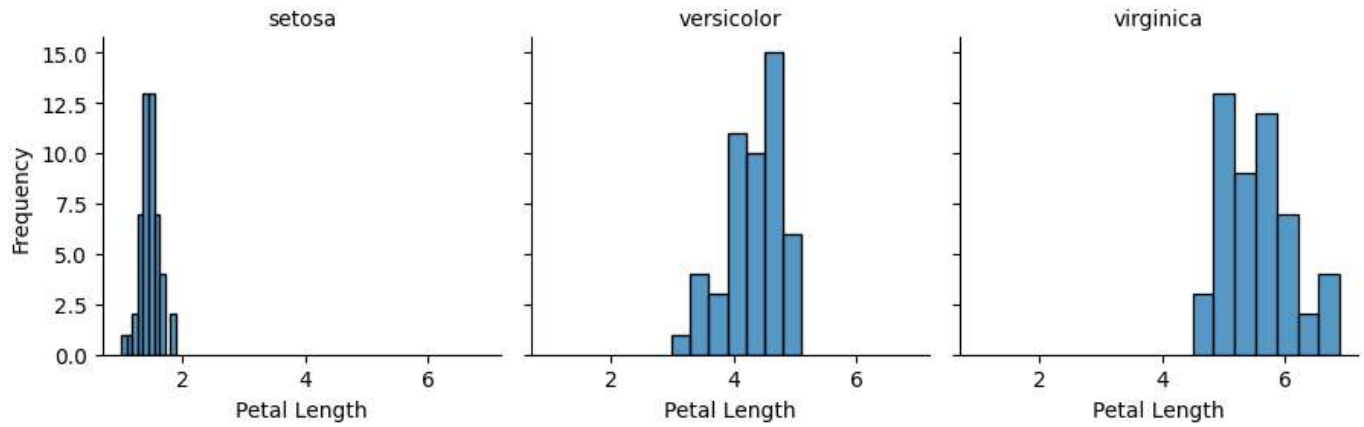
```
# Convert dataset into years vs months matrix
flights_matrix = df.pivot(
    index="year",
    columns="month",
    values="passengers"
)
# Create clustermap
sns.clustermap(
    flights_matrix,
    cmap="Blues",
    annot=True,
    fmt="d"
)
plt.show()
```



Q5. Using the 'iris' dataset, create a FacetGrid that plots a histogram of petal_length separately for each species.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load iris dataset
df = sns.load_dataset("iris")
# Create FacetGrid
g = sns.FacetGrid(
    df,
    col="species"
)
# Plot histogram of petal length
g.map(
    sns.histplot,
    "petal_length"
)
g.set_axis_labels("Petal Length", "Frequency")
g.set_titles("{col_name}")
plt.show()
```

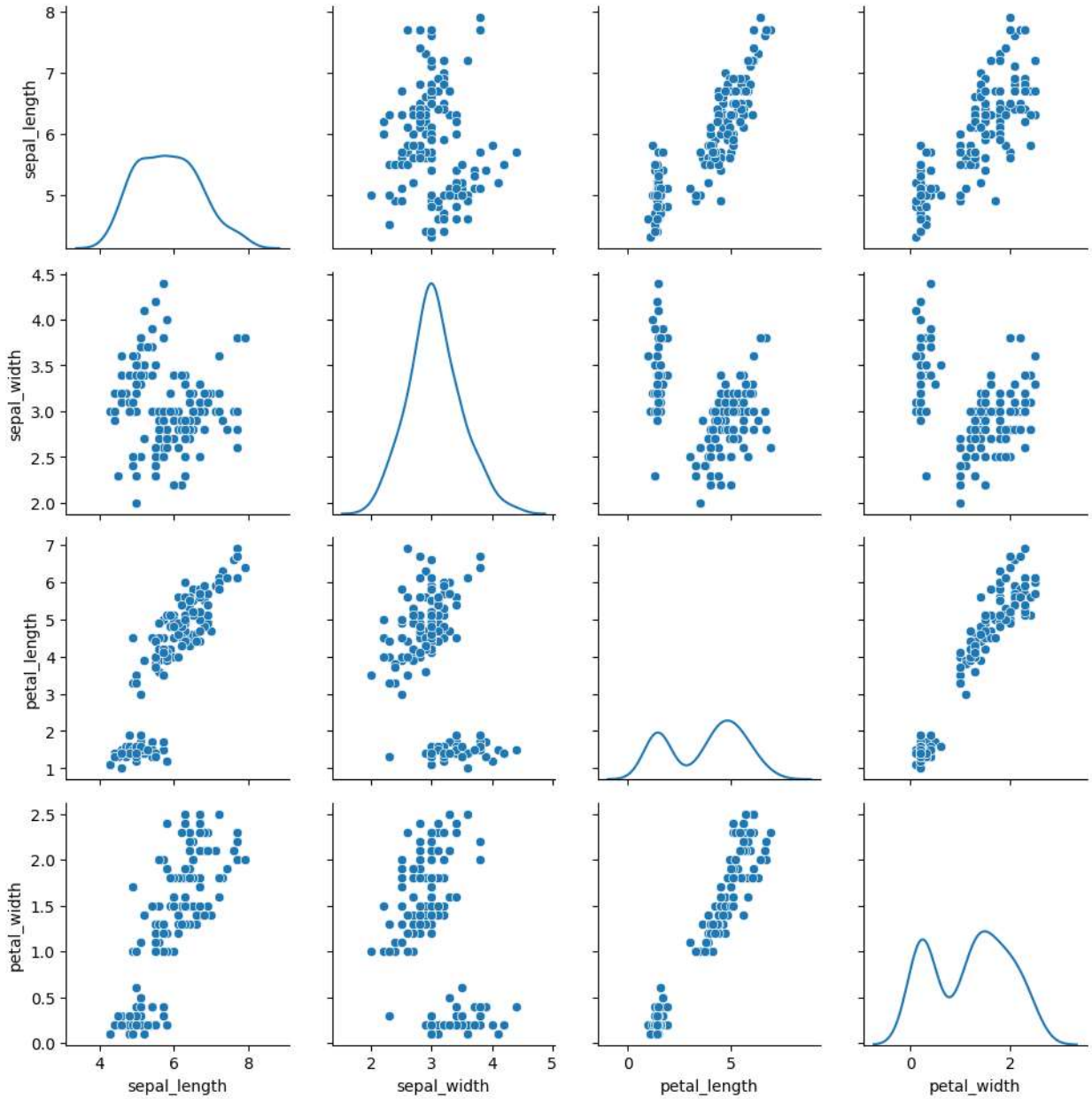


Q6. Using the 'iris' dataset, create a PairGrid where the diagonal shows kde plots and the off-diagonal shows scatter plots for all numeric columns.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load iris dataset
```

```
df = sns.load_dataset("iris")
# Select numeric columns
numeric_df = df.select_dtypes(include="number")
# Create PairGrid
g = sns.PairGrid(numeric_df)
# KDE plots on diagonal
g.map_diag(sns.kdeplot)
# Scatter plots off-diagonal
g.map_offdiag(sns.scatterplot)
plt.show()
```



Q7. Use pairplot() on the 'iris' dataset to generate pairwise scatter plots for all features, with points colored by the species column.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load iris dataset
df = sns.load_dataset("iris")
# Create pairplot
sns.pairplot(
    df,
    hue="species"
)
plt.show()
```

Q8. Write code that applies set_style('darkgrid'), set_context('talk'), and a custom color palette using set_palette(), then plot a simple bar chart to show the effect of these settings.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Apply Seaborn styling
sns.set_style("darkgrid")
sns.set_context("talk")
sns.set_palette("Set2")
# Sample data
products = ["Product A", "Product B", "Product C"]
sales = [25000, 18000, 22000]
# Create bar chart
plt.figure(figsize=(8, 5))
sns.barplot(
    x=products,
    y=sales
)
plt.title("Product Sales")
plt.xlabel("Product")
plt.ylabel("Sales")
plt.show()
```



Q9. Using the 'tips' dataset, create a jointplot() with kind='hex' to visualize the joint distribution of total_bill and tip, along with their individual histograms on the margins.

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load tips dataset
df = sns.load_dataset("tips")
# Create hex joint plot
sns.jointplot(
    data=df,
    x="total_bill",
    y="tip",
    kind="hex"
)
plt.show()
```

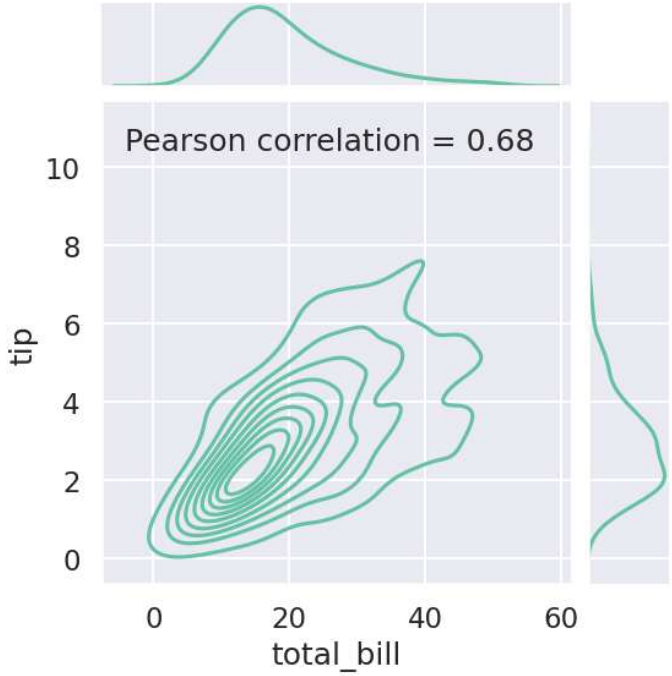
Q10. Using the 'tips' dataset, create a jointplot() with kind='kde' for total_bill and tip, and add a text annotation on the plot showing the Pearson correlation coefficient between the two variables.

Code:

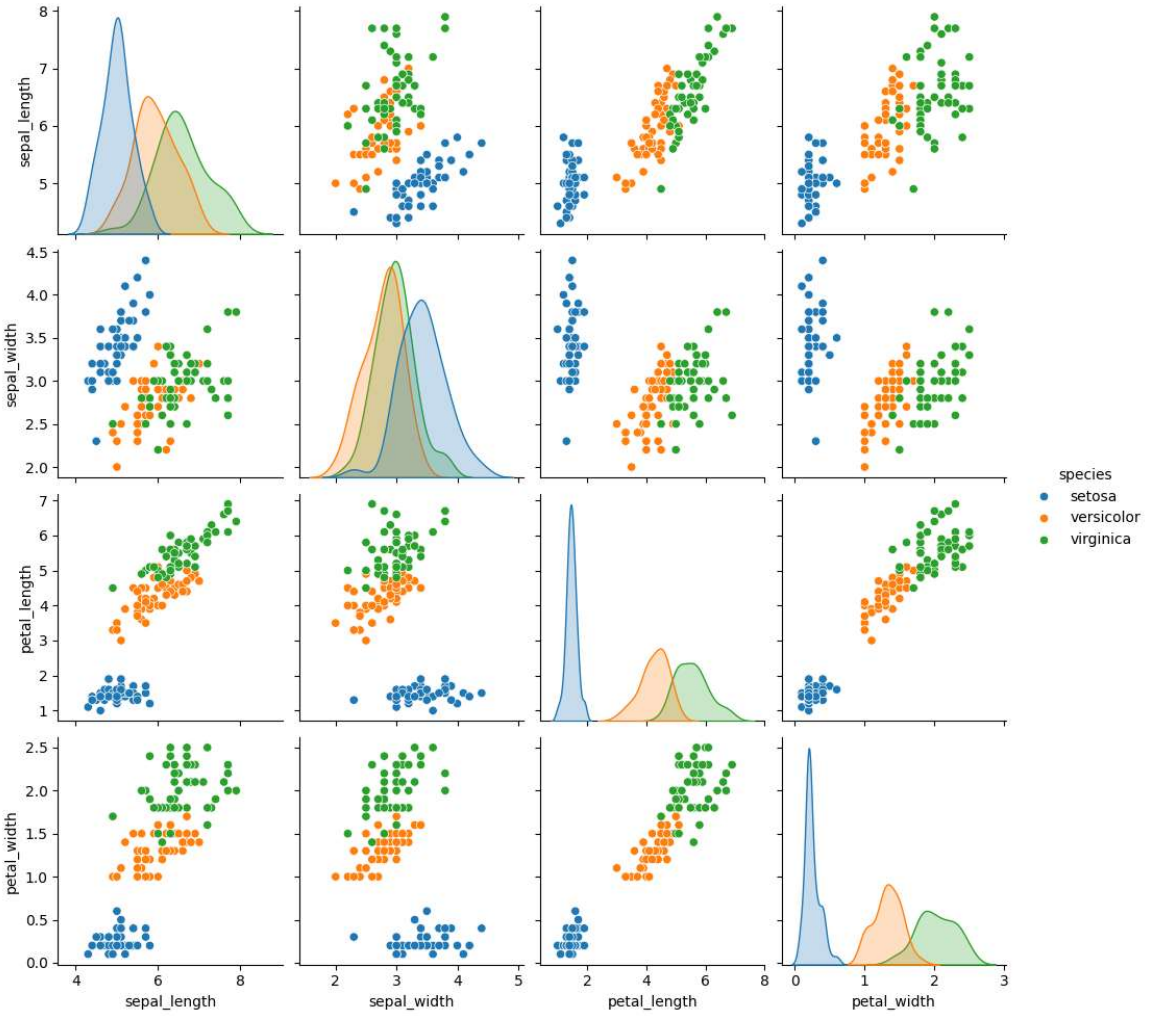
```
import seaborn as sns
import matplotlib.pyplot as plt
# Load tips dataset
df = sns.load_dataset("tips")
# Calculate Pearson correlation coefficient
```

```
correlation = df["total_bill"].corr(df["tip"])
# Create KDE joint plot
g = sns.jointplot(
    data=df,
    x="total_bill",
    y="tip",
    kind="kde"
)
# Add correlation annotation
g.ax_joint.text(
    0.05,
    0.95,
    f"Pearson correlation = {correlation:.2f}",
    transform=g.ax_joint.transAxes,
    ha="left",
    va="top"

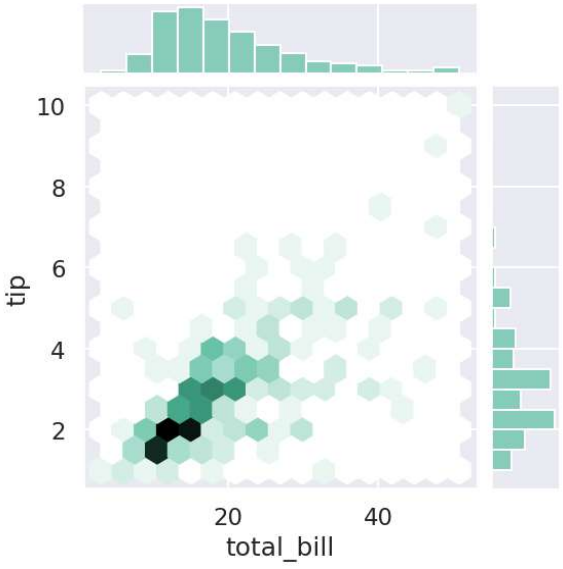
plt.show()
```



Q7:


sepal_length
sepal_width
petal_length
petal_width
species
setosa
versicolor
virginica

Q9:


tip
total_bill